

Quadratic Equation and Function, group *Alpha α* -i

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $-9x^2 + 4x - 2 = 0$.. ??? .. 2

(b) $-8x^2 + 3x - 2 = 0$.. ??? .. 0

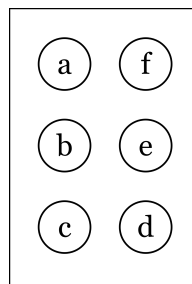
(c) $f(x) = -7x^2 + 4x + 2$??? $2/7$

(d) $f(x) = -3x^2 + 8x + 4$??? $22/3$

(e) $-3x^2 - 18x - 24 = 0$. ??? . -6

(f) $2x^2 - 2x - 40 = 0$.. ??? .. 1

1.



Braille letter

Quadratic Equation and Function, group *Alpha α* -ii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $-7x^2 + 4x - 1 = 0$.. ??? .. 0

(b) $x^2 + 5x + 3 = 0$... ??? ... 0

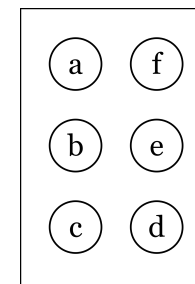
(c) $f(x) = 6x^2 - 8x - 3$??? $2/3$

(d) $f(x) = -4x^2 - 4x + 7$??? $9/2$

(e) $-x^2 - x + 2 = 0$... ??? ... -1

(f) $20x^2 - 6x - 2 = 0$??? $-7/10$

2.



Braille letter

Quadratic Equation and Function, group *Alpha α* -iii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $4x^2 - x - 4 = 0$??? 2

(b) $-3x^2 - 4x - 1 = 0$.. ??? .. 2

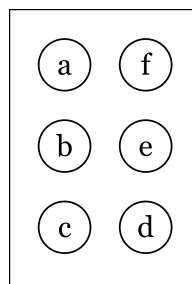
(c) $f(x) = -7x^2 + 4x + 6$??? $2/7$

(d) $f(x) = -2x^2 - x - 3$??? $-23/8$

(e) $7x^2 + 21x + 14 = 0$.. ??? .. -3

(f) $-24x^2 - 10x - 1 = 0$??? $-1/12$

3.



Braille letter

Quadratic Equation and Function, group *Alpha α* -iv

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $-8x^2 + 3x - 5 = 0$. ??? . 0

(b) $-2x^2 + 3x + 1 = 0$. ??? . 0

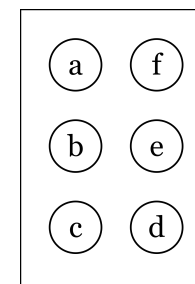
(c) $f(x) = x^2 + x + 5$??? $-1/2$

(d) $f(x) = x^2 + 2x + 4$. ??? . 1

(e) $3x^2 + 15x + 12 = 0$??? -5

(f) $-4x^2 - 2x + 2 = 0$??? $-1/2$

4.



Braille letter

Quadratic Equation and Function, group *Beta* β -i

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $x^2 - x + 7 = 0$??? 0

(b) $-5x^2 + 3x - 6 = 0$.. ??? .. 1

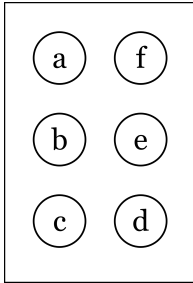
(c) $f(x) = -4x^2 - 2x + 5$??? $-1/4$

(d) $f(x) = -x^2 + 8x + 6$. ??? . 22

(e) $-6x^2 - 6x + 36 = 0$.. ??? .. 0

(f) $2x^2 - x - 6 = 0$... ??? ... $-7/2$

1.



Braille letter

Quadratic Equation and Function, group *Beta* β -ii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $6x^2 + 5x + 9 = 0$... ??? ... 0

(b) $2x^2 + 2x + 6 = 0$... ??? ... 2

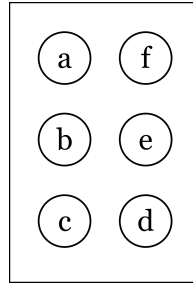
(c) $f(x) = 4x^2 + 8x - 6$??? -1

(d) $f(x) = -7x^2 + x + 3$??? $43/28$

(e) $x^2 + 4x - 32 = 0$... ??? ... -4

(f) $-6x^2 - x + 12 = 0$. ??? . $-1/6$

2.



Braille letter

Quadratic Equation and Function, group *Beta* β -iii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $3x^2 + 3x - 2 = 0$.. ??? .. 1

(b) $-3x^2 + 7x - 1 = 0$.. ??? .. 2

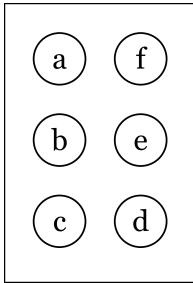
(c) $f(x) = 6x^2 - 2x - 4$??? $-1/6$

(d) $f(x) = x^2 + x + 2$. ??? . $3/4$

(e) $-2x^2 + 14x - 20 = 0$??? 6

(f) $-10x^2 + 12x - 2 = 0$??? $6/5$

3.



Braille letter

Quadratic Equation and Function, group *Beta* β -iv

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-2x^2 + 2x - 1 = 0$.. ??? .. 1

(b) $3x^2 + x - 3 = 0$??? 2

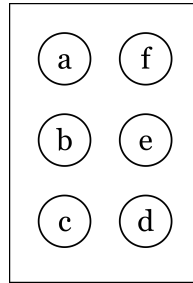
(c) $f(x) = 2x^2 + 3x + 8$??? $-3/4$

(d) $f(x) = -2x^2 + 5x + 1$??? $29/8$

(e) $-2x^2 - 22x - 56 = 0$??? -11

(f) $-4x^2 + 5x - 1 = 0$.. ??? .. $5/4$

4.



Braille letter

Quadratic Equation and Function, group *Gamma* γ -i

Quadratic Equation and Function, group *Gamma* γ -ii

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $-4x^2 - 3x + 5 = 0$.. ??? .. 2

(b) $8x^2 + 7x - 6 = 0$... ??? ... 2

(c) $f(x) = 5x^2 - 3x + 2$. ??? . $3/10$

(d) $f(x) = 3x^2 - 5x + 1$??? $-13/12$

(e) $x^2 - 10x + 24 = 0$.. ??? .. 10

(f) $-2x^2 - 11x - 15 = 0$. ??? . $1/2$

1.

a

f

b

e

c

d

Braille letter

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $x^2 - 7x + 5 = 0$... ??? ... 2

(b) $-5x^2 - 5x + 5 = 0$.. ??? .. 0

(c) $f(x) = 2x^2 - 5x - 6$??? $-5/4$

(d) $f(x) = 8x^2 + 4x + 1$. ??? . 0

(e) $-3x^2 + 6x + 9 = 0$.. ??? .. 5

(f) $-15x^2 + 7x + 2 = 0$??? $13/15$

2.

a

f

b

e

c

d

Braille letter

Quadratic Equation and Function, group *Gamma* γ -iii

Quadratic Equation and Function, group *Gamma* γ -iv

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $x^2 - 5x + 6 = 0$??? 2

(b) $-4x^2 + 4x + 1 = 0$... ??? ... 2

(c) $f(x) = -4x^2 + 2x + 8$??? $1/4$

(d) $f(x) = -4x^2 + x - 2$??? $-15/16$

(e) $-x^2 - 4x - 3 = 0$... ??? ... -2

(f) $3x^2 + 13x + 12 = 0$.. ??? .. $5/3$

3.

a

f

b

e

c

d

Braille letter

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $3x^2 - 3x - 5 = 0$... ??? ... 2

(b) $x^2 + 6x + 4 = 0$... ??? ... 2

(c) $f(x) = x^2 + 4x + 6$. ??? . -2

(d) $f(x) = x^2 + 5x - 6$??? $-37/4$

(e) $-x^2 + 2x + 35 = 0$.. ??? .. -1

(f) $-10x^2 - 18x - 8 = 0$??? $-1/5$

4.

a

f

b

e

c

d

Braille letter

Quadratic Equation and Function, group *Delta δ* -i

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-6x^2 + 2x - 3 = 0$.. ??? .. 0

(b) $4x^2 + 4x + 8 = 0$... ??? ... 0

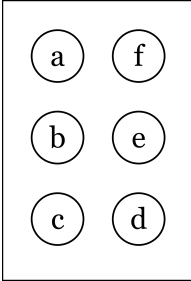
(c) $f(x) = -4x^2 + 3x - 7$??? $-3/8$

(d) $f(x) = 6x^2 - 8x - 2$??? $-11/3$

(e) $-5x^2 + 10x + 15 = 0$. ??? . 1

(f) $-8x^2 - 6x + 2 = 0$. ??? . $-3/4$

1.



Braille letter

Quadratic Equation and Function, group *Delta δ* -ii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-6x^2 - 3x - 3 = 0$.. ??? .. 0

(b) $-2x^2 - 5x - 5 = 0$.. ??? .. 2

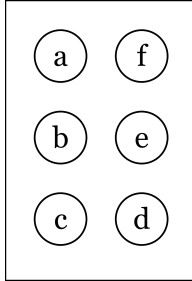
(c) $f(x) = -5x^2 - 3x + 3$??? $3/10$

(d) $f(x) = 2x^2 - 7x + 6$??? $-25/8$

(e) $-7x^2 - 14x + 21 = 0$. ??? . 1

(f) $4x^2 - 7x + 3 = 0$.. ??? .. $1/4$

2.



Braille letter

Quadratic Equation and Function, group *Delta δ* -iii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $2x^2 + 5x + 7 = 0$... ??? ... 0

(b) $4x^2 - 4x + 5 = 0$... ??? ... 0

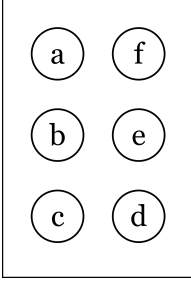
(c) $f(x) = -7x^2 - 2x + 4$??? $-1/7$

(d) $f(x) = 2x^2 + 5x + 6$??? $-1/8$

(e) $-2x^2 + 14x - 12 = 0$. ??? . 7

(f) $2x^2 - 2x - 4 = 0$.. ??? .. -3

3.



Braille letter

Quadratic Equation and Function, group *Delta δ* -iv

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $x^2 + 3x - 2 = 0$... ??? ... 2

(b) $x^2 - 6x - 4 = 0$... ??? ... 0

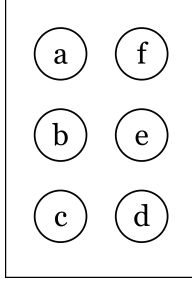
(c) $f(x) = 5x^2 + 2x + 2$??? $-1/5$

(d) $f(x) = x^2 - 4x + 2$??? -3

(e) $-x^2 + 9x - 14 = 0$. ??? . 6

(f) $6x^2 - 5x - 6 = 0$.. ??? .. $5/6$

4.



Braille letter

Quadratic Equation and Function, group *Epsilon* ϵ -i

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $-7x^2 - 6x - 5 = 0$.. ??? .. 0

(b) $8x^2 + x + 6 = 0$... ??? ... 1

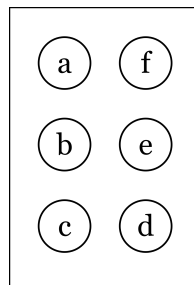
(c) $f(x) = -7x^2 - x + 6$??? $1/14$

(d) $f(x) = 7x^2 - 5x + 3$??? $17/28$

(e) $-6x^2 + 30x - 24 = 0$. ??? . 5

(f) $-12x^2 - 14x - 4 = 0$??? $1/6$

1.



Braille letter

Quadratic Equation and Function, group *Epsilon* ϵ -ii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $-3x^2 + 6x - 4 = 0$. ??? . 0

(b) $-6x^2 - 6x + 2 = 0$. ??? . 0

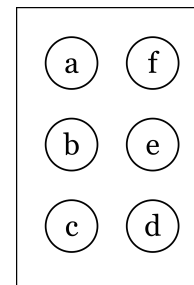
(c) $f(x) = 6x^2 + 8x - 5$??? $2/3$

(d) $f(x) = x^2 + x + 2$. ??? . $3/4$

(e) $-6x^2 - 18x + 24 = 0$??? -6

(f) $x^2 - 10x + 24 = 0$??? -2

2.



Braille letter

Quadratic Equation and Function, group *Epsilon* ϵ -iii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $9x^2 + 4x - 2 = 0$... ??? ... 1

(b) $-6x^2 - 6x - 6 = 0$.. ??? .. 0

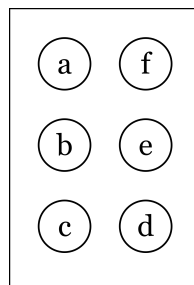
(c) $f(x) = -5x^2 + 7x + 9$??? $7/10$

(d) $f(x) = x^2 - 3x - 3$??? $-15/4$

(e) $2x^2 - 2x - 4 = 0$... ??? ... 3

(f) $x^2 + 5x + 4 = 0$... ??? ... -5

3.



Braille letter

Quadratic Equation and Function, group *Epsilon* ϵ -iv

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $9x^2 - 4x + 1 = 0$... ??? ... 1

(b) $5x^2 - 3x + 2 = 0$... ??? ... 0

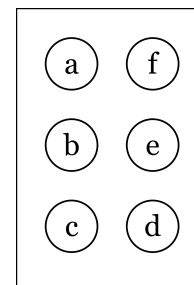
(c) $f(x) = -3x^2 - 9x - 4$??? $-3/2$

(d) $f(x) = -x^2 - 3x - 8$??? $-7/4$

(e) $2x^2 + 2x - 84 = 0$.. ??? .. -1

(f) $-2x^2 + 6x - 4 = 0$.. ??? .. 3

4.



Braille letter

Quadratic Equation and Function, group *Zeta* ζ -i

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $x^2 + 2x - 2 = 0$??? 2

(b) $9x^2 - x - 6 = 0$??? 1

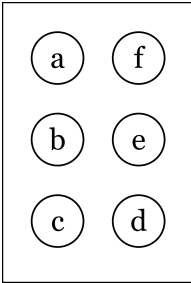
(c) $f(x) = -5x^2 + 9x + 3$??? 9/10

(d) $f(x) = -x^2 - x - 1$??? -1/4

(e) $6x^2 - 36x + 48 = 0$.. ??? .. 5

(f) $4x^2 - 3x - 1 = 0$.. ??? .. 3/4

1.



Braille letter

Quadratic Equation and Function, group *Zeta* ζ -ii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-2x^2 - 3x + 7 = 0$.. ??? .. 0

(b) $-2x^2 + 7x - 8 = 0$.. ??? .. 0

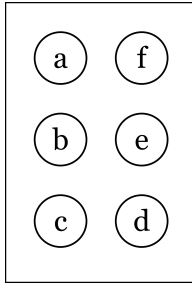
(c) $f(x) = -3x^2 - 8x - 7$??? 4/3

(d) $f(x) = -x^2 + 7x + 4$??? 57/4

(e) $3x^2 + 27x + 60 = 0$. ??? . -8

(f) $-3x^2 - 5x + 2 = 0$. ??? . -5/3

2.



Braille letter

Quadratic Equation and Function, group *Zeta* ζ -iii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $6x^2 - 3x + 3 = 0$... ??? ... 0

(b) $-x^2 - 2x + 1 = 0$.. ??? .. 0

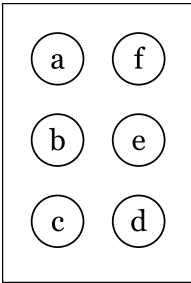
(c) $f(x) = -4x^2 + 7x - 4$??? 7/8

(d) $f(x) = 8x^2 - 6x + 7$??? 19/8

(e) $-3x^2 - 9x + 12 = 0$. ??? . -3

(f) $-30x^2 - 4x + 2 = 0$??? -2/15

3.



Braille letter

Quadratic Equation and Function, group *Zeta* ζ -iv

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-x^2 - 3x + 7 = 0$. ??? . 2

(b) $-4x^2 + 8x - 3 = 0$. ??? . 1

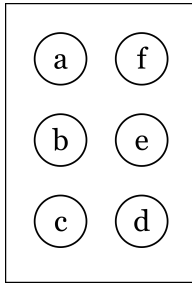
(c) $f(x) = -x^2 - 2x + 3$??? 1

(d) $f(x) = 4x^2 + x + 1$??? 7/16

(e) $7x^2 + 21x + 14 = 0$??? -3

(f) $2x^2 + 4x - 6 = 0$. ??? . -2

4.



Braille letter

Quadratic Equation and Function, group *Eta η* -i

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $x^2 + 2x + 1 = 0$??? 0

(b) $-5x^2 + 6x + 8 = 0$.. ??? .. 2

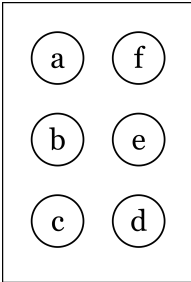
(c) $f(x) = -2x^2 - 8x - 1$??? -2

(d) $f(x) = 3x^2 - 3x + 8$??? 13/4

(e) $-x^2 + 3x - 2 = 0$... ??? ... 5

(f) $-4x^2 + 2x + 20 = 0$. ??? . 1/2

1.



Braille letter

Quadratic Equation and Function, group *Eta η* -ii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $3x^2 + 7x + 6 = 0$... ??? ... 1

(b) $3x^2 + 3x + 9 = 0$... ??? ... 0

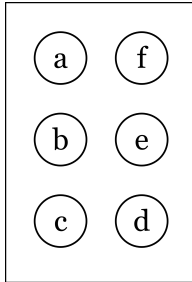
(c) $f(x) = 9x^2 + x + 2$. ??? . 1/18

(d) $f(x) = -5x^2 + 8x - 1$??? 27/10

(e) $5x^2 + 25x + 20 = 0$.. ??? .. -8

(f) $4x^2 - 10x + 6 = 0$.. ??? .. 5/2

2.



Braille letter

Quadratic Equation and Function, group *Eta η* -iii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-8x^2 - 5x - 2 = 0$.. ??? .. 0

(b) $-x^2 - 4x + 2 = 0$.. ??? .. 1

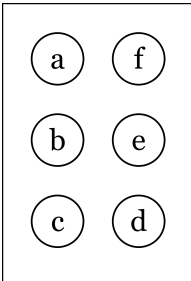
(c) $f(x) = -8x^2 + x + 1$??? 1/16

(d) $f(x) = x^2 - 4x - 5$. ??? . -9

(e) $x^2 - 5x - 24 = 0$... ??? ... 5

(f) $-24x^2 - 14x - 2 = 0$??? 1/12

3.



Braille letter

Quadratic Equation and Function, group *Eta η* -iv

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $3x^2 + x - 6 = 0$... ??? ... 2

(b) $9x^2 - 5x - 1 = 0$... ??? ... 0

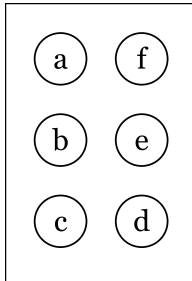
(c) $f(x) = 5x^2 + 2x + 3$??? 1/5

(d) $f(x) = 7x^2 - 8x + 7$??? 17/14

(e) $-3x^2 - 27x - 60 = 0$??? -9

(f) $12x^2 - 2x - 2 = 0$. ??? . -5/6

4.



Braille letter

Quadratic Equation and Function, group *Theta* θ -i

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $5x^2 - x - 2 = 0$??? 2

(b) $-4x^2 - 7x - 5 = 0$... ??? ... 2

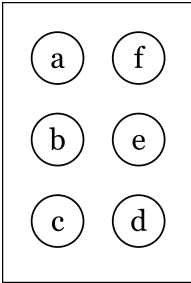
(c) $f(x) = 8x^2 + 4x + 7$. ??? . $1/4$

(d) $f(x) = -4x^2 + x - 8$??? $-63/16$

(e) $2x^2 - 14x + 12 = 0$.. ??? .. 7

(f) $-2x^2 + 10x - 12 = 0$.. ??? .. 5

1.



Braille letter

Quadratic Equation and Function, group *Theta* θ -ii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $4x^2 + 6x + 5 = 0$... ??? ... 1

(b) $6x^2 + 8x - 9 = 0$... ??? ... 2

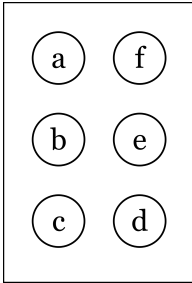
(c) $f(x) = -3x^2 - 5x - 3$??? $5/6$

(d) $f(x) = -x^2 - 3x + 9$??? $27/4$

(e) $-x^2 - 7x - 12 = 0$. ??? . -4

(f) $-3x^2 - 3x + 18 = 0$??? -1

2.



Braille letter

Quadratic Equation and Function, group *Theta* θ -iii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $2x^2 - 4x - 3 = 0$... ??? ... 2

(b) $2x^2 + x - 4 = 0$??? 2

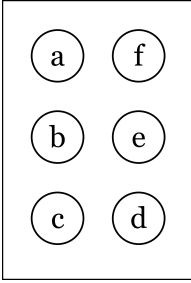
(c) $f(x) = -4x^2 + 9x + 1$??? $9/8$

(d) $f(x) = -3x^2 + 8x - 1$??? $29/6$

(e) $7x^2 - 28x + 21 = 0$.. ??? .. 4

(f) $4x^2 - 3x - 1 = 0$.. ??? .. $5/4$

3.



Braille letter

Quadratic Equation and Function, group *Theta* θ -iv

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-2x^2 - 2x + 4 = 0$... ??? ... 1

(b) $-4x^2 - 3x + 2 = 0$... ??? ... 2

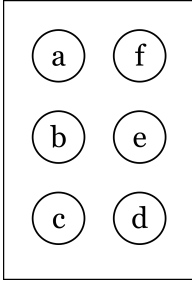
(c) $f(x) = x^2 - 4x + 3$... ??? ... 2

(d) $f(x) = -5x^2 + 4x - 6$??? $-11/5$

(e) $7x^2 - 14x - 21 = 0$... ??? ... 2

(f) $-3x^2 + 2x + 1 = 0$.. ??? .. $2/3$

4.



Braille letter

Quadratic Equation and Function, group *Iota* ι -i

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $3x^2 + x - 1 = 0$??? 2

(b) $-3x^2 - 9x + 2 = 0$.. ??? .. 2

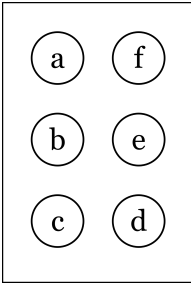
(c) $f(x) = 6x^2 + 7x + 8$. ??? . 7/12

(d) $f(x) = -4x^2 + 2x - 5$??? -9/4

(e) $-4x^2 - 8x + 32 = 0$. ??? . -1

(f) $4x^2 + 8x + 3 = 0$.. ??? .. -1

1.



Braille letter

Quadratic Equation and Function, group *Iota* ι -ii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $3x^2 + 2x - 4 = 0$... ??? ... 2

(b) $4x^2 - 2x + 4 = 0$... ??? ... 1

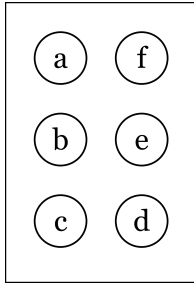
(c) $f(x) = 7x^2 - 5x - 3$??? 5/14

(d) $f(x) = 6x^2 + 9x + 4$??? -11/8

(e) $3x^2 + 27x + 60 = 0$. ??? . -9

(f) $-2x^2 - 7x - 6 = 0$. ??? . -1/2

2.



Braille letter

Quadratic Equation and Function, group *Iota* ι -iii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-5x^2 + 5x - 2 = 0$.. ??? .. 1

(b) $x^2 + 4x - 7 = 0$??? 2

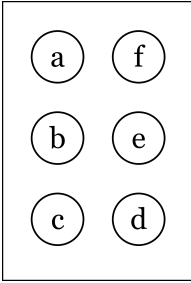
(c) $f(x) = -6x^2 + 4x - 4$??? 1/3

(d) $f(x) = -9x^2 + 4x + 4$??? 22/9

(e) $x^2 + 2x - 15 = 0$... ??? ... -1

(f) $4x^2 + 6x - 10 = 0$. ??? . -3/2

3.



Braille letter

Quadratic Equation and Function, group *Iota* ι -iv

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-2x^2 - 6x - 7 = 0$... ??? ... 2

(b) $6x^2 + 2x - 1 = 0$??? 2

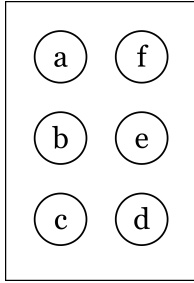
(c) $f(x) = -4x^2 + 3x - 9$. ??? . 3/8

(d) $f(x) = -8x^2 + 3x + 8$??? 137/32

(e) $-2x^2 - 2x + 40 = 0$.. ??? .. -3

(f) $-6x^2 + 2x + 8 = 0$.. ??? .. 1/3

4.



Braille letter

Quadratic Equation and Function, group *Kappa* κ -i

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $7x^2 + 3x - 2 = 0$... ??? ... 2

(b) $4x^2 + 7x + 8 = 0$... ??? ... 0

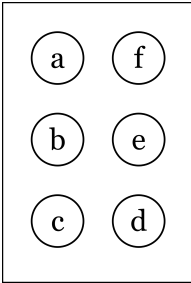
(c) $f(x) = -3x^2 - 8x + 3$??? $4/3$

(d) $f(x) = 6x^2 - 3x + 7$??? $25/8$

(e) $-3x^2 + 15x - 12 = 0$. ??? . 5

(f) $2x^2 + x - 28 = 0$. ??? . $-1/2$

1.



Braille letter

Quadratic Equation and Function, group *Kappa* κ -ii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-3x^2 + 2x + 3 = 0$.. ??? .. 2

(b) $x^2 + 2x + 1 = 0$??? 2

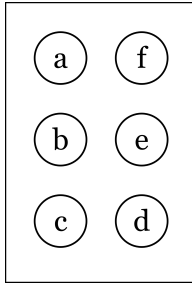
(c) $f(x) = -2x^2 + 2x - 1$??? $-1/2$

(d) $f(x) = -3x^2 + 3x - 1$??? $1/4$

(e) $-x^2 - 9x - 8 = 0$... ??? ... -8

(f) $8x^2 + 15x + 7 = 0$.. ??? .. $1/8$

2.



Braille letter

Quadratic Equation and Function, group *Kappa* κ -iii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $2x^2 - 2x - 1 = 0$... ??? ... 1

(b) $-x^2 - 6x + 2 = 0$.. ??? .. 2

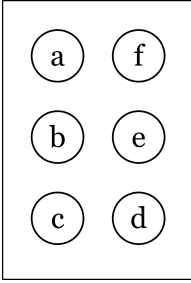
(c) $f(x) = 3x^2 + 5x + 7$??? $-5/6$

(d) $f(x) = -3x^2 + 6x + 3$??? $9/2$

(e) $-x^2 - 12x - 35 = 0$??? -12

(f) $4x^2 - 4x - 8 = 0$... ??? ... 1

3.



Braille letter

Quadratic Equation and Function, group *Kappa* κ -iv

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-x^2 + 4x - 2 = 0$... ??? ... 2

(b) $-x^2 + 4x - 6 = 0$... ??? ... 2

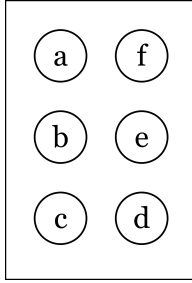
(c) $f(x) = -8x^2 + 7x - 2$??? $-7/16$

(d) $f(x) = 6x^2 - 5x - 2$??? $-49/24$

(e) $4x^2 - 12x - 40 = 0$... ??? ... 3

(f) $2x^2 + x - 3 = 0$... ??? ... $-5/2$

4.



Braille letter

Quadratic Equation and Function, group *Lambda* λ -i

Quadratic Equation and Function, group *Lambda* λ -ii

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $-x^2 - x - 5 = 0$... ??? ... 0

(b) $x^2 - 2x + 8 = 0$??? 0

(c) $f(x) = -3x^2 + 8x + 1$??? $-4/3$

(d) $f(x) = x^2 - 3x + 7$.. ??? .. $5/4$

(e) $2x^2 + 20x + 48 = 0$.. ??? .. -7

(f) $3x^2 + 9x + 6 = 0$.. ??? .. -3

1.

a

b

c

f

e

d

Braille letter

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $x^2 + 3x - 1 = 0$??? 0

(b) $-x^2 - 3x + 6 = 0$... ??? ... 2

(c) $f(x) = x^2 - 8x - 5$. ??? . -4

(d) $f(x) = -4x^2 - 5x + 4$??? $57/16$

(e) $x^2 - 10x + 24 = 0$.. ??? .. 12

(f) $8x^2 - 2x - 3 = 0$... ??? ... $1/4$

2.

a

b

c

f

e

d

Braille letter

Quadratic Equation and Function, group *Lambda* λ -iii

Quadratic Equation and Function, group *Lambda* λ -iv

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $-7x^2 + 3x - 2 = 0$.. ??? .. 0

(b) $-3x^2 - 4x - 3 = 0$.. ??? .. 0

(c) $f(x) = -9x^2 + x - 5$??? $1/18$

(d) $f(x) = 4x^2 + 4x - 4$. ??? . -5

(e) $4x^2 - 36x + 80 = 0$.. ??? .. 10

(f) $-12x^2 - 17x - 6 = 0$??? $-1/12$

3.

a

b

c

f

e

d

Braille letter

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $2x^2 + 5x + 3 = 0$.. ??? .. 2

(b) $8x^2 + 2x - 7 = 0$.. ??? .. 0

(c) $f(x) = 6x^2 + 3x + 2$??? $1/4$

(d) $f(x) = x^2 + 5x - 2$??? $-29/4$

(e) $-5x^2 - 5x + 10 = 0$??? -1

(f) $3x^2 + 5x + 2 = 0$. ??? . $-1/3$

4.

a

b

c

f

e

d

Braille letter

Quadratic Equation and Function, group *Mu* μ -i

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $3x^2 - 3x + 9 = 0$.. ??? .. 0

(b) $7x^2 - 4x - 2 = 0$.. ??? .. 2

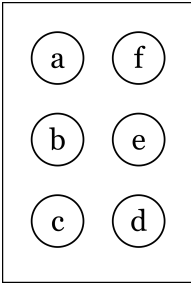
(c) $f(x) = 2x^2 + 9x - 2$??? $-9/4$

(d) $f(x) = x^2 - 3x + 1$??? $-5/4$

(e) $x^2 - 3x + 2 = 0$... ??? ... 3

(f) $-2x^2 - 5x + 3 = 0$. ??? . $7/2$

1.



Braille letter

Quadratic Equation and Function, group *Mu* μ -ii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-x^2 - 2x - 4 = 0$??? 0

(b) $2x^2 + x - 1 = 0$??? 1

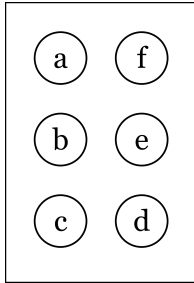
(c) $f(x) = -5x^2 - x - 5$??? $-1/10$

(d) $f(x) = -4x^2 - 7x + 8$??? $113/16$

(e) $-4x^2 + 8x + 12 = 0$.. ??? .. 2

(f) $6x^2 + 14x + 8 = 0$.. ??? .. $-1/3$

2.



Braille letter

Quadratic Equation and Function, group *Mu* μ -iii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $x^2 + 7x + 5 = 0$... ??? ... 2

(b) $-4x^2 - 2x - 7 = 0$.. ??? .. 2

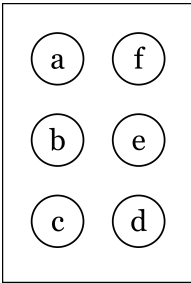
(c) $f(x) = 2x^2 - 2x - 2$??? $1/2$

(d) $f(x) = -x^2 + 5x + 4$??? $33/4$

(e) $-x^2 + 2x + 8 = 0$.. ??? .. 2

(f) $24x^2 - 10x + 1 = 0$??? $-1/12$

3.



Braille letter

Quadratic Equation and Function, group *Mu* μ -iv

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $2x^2 + 4x - 7 = 0$.. ??? .. 2

(b) $-9x^2 + x + 2 = 0$.. ??? .. 1

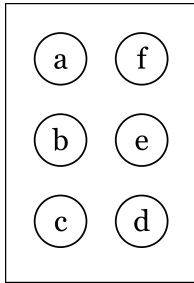
(c) $f(x) = -x^2 + 2x - 7$??? -1

(d) $f(x) = -x^2 + 3x + 5$??? $19/4$

(e) $2x^2 + 12x + 10 = 0$. ??? . -6

(f) $3x^2 - 4x - 4 = 0$.. ??? .. $4/3$

4.



Braille letter

Quadratic Equation and Function, group $Nu \nu$ -i

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $-5x^2 - 2x - 3 = 0$.. ??? .. 0

(b) $-3x^2 - x - 1 = 0$.. ??? .. 0

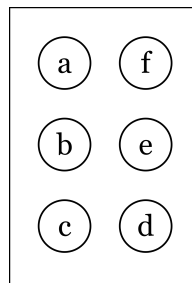
(c) $f(x) = 5x^2 - 7x - 1$??? $7/10$

(d) $f(x) = 3x^2 - x - 3$??? $-19/12$

(e) $x^2 - 3x + 2 = 0$... ??? ... 3

(f) $2x^2 - 6x + 4 = 0$... ??? ... 1

1.



Braille letter

Quadratic Equation and Function, group $Nu \nu$ -ii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $-x^2 + x + 2 = 0$... ??? ... 2

(b) $6x^2 + x - 1 = 0$??? 0

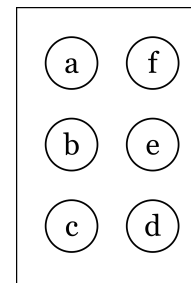
(c) $f(x) = 9x^2 - 2x - 2$. ??? . $1/9$

(d) $f(x) = 8x^2 + 2x + 1$. ??? . $3/8$

(e) $-3x^2 + 3x + 18 = 0$.. ??? .. 1

(f) $-14x^2 + 3x + 5 = 0$??? $-17/14$

2.



Braille letter

Quadratic Equation and Function, group $Nu \nu$ -iii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $2x^2 - 7x + 8 = 0$... ??? ... 0

(b) $3x^2 + 2x - 3 = 0$... ??? ... 0

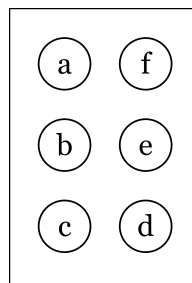
(c) $f(x) = -2x^2 - 3x + 6$??? $3/4$

(d) $f(x) = 3x^2 + x + 8$??? $47/12$

(e) $-x^2 - 7x - 12 = 0$. ??? . -6

(f) $6x^2 + 15x + 9 = 0$. ??? . $1/2$

3.



Braille letter

Quadratic Equation and Function, group $Nu \nu$ -iv

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $8x^2 - 6x + 4 = 0$... ??? ... 0

(b) $-2x^2 + 6x + 1 = 0$.. ??? .. 0

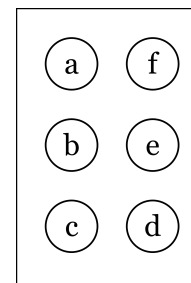
(c) $f(x) = 8x^2 - x - 7$??? $-1/16$

(d) $f(x) = 2x^2 - 8x - 3$??? $-19/2$

(e) $-5x^2 - 30x - 40 = 0$??? -6

(f) $7x^2 + 4x - 3 = 0$.. ??? .. $-4/7$

4.



Braille letter

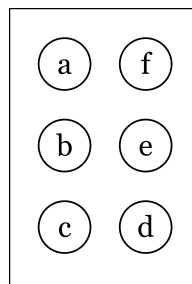
Quadratic Equation and Function, group Xi ξ -i

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $-2x^2 - 7x + 1 = 0$??? 2
- (b) $-6x^2 - 5x - 3 = 0$??? 0
- (c) $f(x) = 7x^2 + x + 2$.. ??? .. $1/14$
- (d) $f(x) = -9x^2 + 5x - 7$??? $-101/36$
- (e) $5x^2 + 20x + 15 = 0$... ??? ... -4
- (f) $-12x^2 + 2x + 4 = 0$.. ??? .. $1/6$

1.



Braille letter

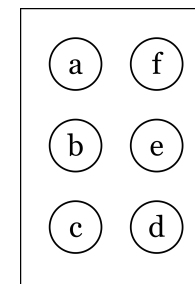
Quadratic Equation and Function, group Xi ξ -ii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $-9x^2 - 9x + 3 = 0$.. ??? .. 2
- (b) $-9x^2 - 2x + 7 = 0$.. ??? .. 2
- (c) $f(x) = x^2 - 2x + 1$.. ??? .. 1
- (d) $f(x) = -7x^2 - 4x - 2$??? $-3/7$
- (e) $x^2 - 4x - 12 = 0$... ??? ... 4
- (f) $-12x^2 - 5x + 3 = 0$??? $13/12$

2.



Braille letter

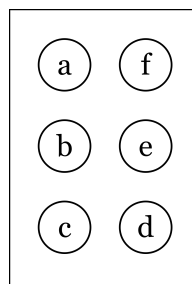
Quadratic Equation and Function, group Xi ξ -iii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $-x^2 + 7x + 4 = 0$... ??? ... 2
- (b) $-x^2 + x + 2 = 0$??? 1
- (c) $f(x) = -3x^2 + 7x - 1$??? $-7/6$
- (d) $f(x) = 2x^2 + x + 6$. ??? . $23/8$
- (e) $3x^2 - 6x - 45 = 0$... ??? ... 3
- (f) $-2x^2 + 2x + 12 = 0$.. ??? .. 5

3.



Braille letter

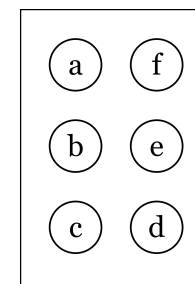
Quadratic Equation and Function, group Xi ξ -iv

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $-5x^2 + 4x + 2 = 0$... ??? ... 2
- (b) $3x^2 + 4x - 2 = 0$... ??? ... 0
- (c) $f(x) = x^2 + 6x - 1$. ??? . -3
- (d) $f(x) = -5x^2 - 3x + 3$??? $69/20$
- (e) $x^2 + 2x - 35 = 0$... ??? ... -2
- (f) $-4x^2 + 2x + 2 = 0$.. ??? .. $1/2$

4.



Braille letter

Quadratic Equation and Function, group *Omicron o* -i

Quadratic Equation and Function, group *Omicron o* -ii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a)

$3x^2 + 5x + 9 = 0$

...

???

...

0
- (b)

$x^2 - 4x + 6 = 0$

....

???

....

2
- (c)

$f(x) = 3x^2 - 5x + 4$

???

$-5/6$
- (d)

$f(x) = 3x^2 - 7x - 6$

???

$-85/12$
- (e)

$-x^2 - 14x - 45 = 0$

.

???

.

-17
- (f)

$-6x^2 + 2x + 4 = 0$

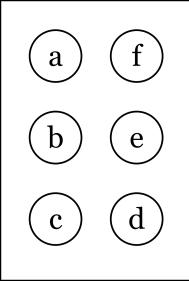
..

???

..

$1/3$

1.



Braille letter

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a)

$-4x^2 - 8x + 2 = 0$

..

???

..

2
- (b)

$-2x^2 + 9x + 2 = 0$

..

???

..

2
- (c)

$f(x) = 5x^2 + 3x + 6$

???

$-3/10$
- (d)

$f(x) = -4x^2 + x + 7$

???

$57/16$
- (e)

$-4x^2 - 8x + 32 = 0$

.

???

.

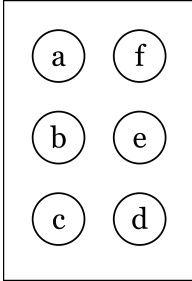
-1
- (f)

$-3x^2 + 10x - 8 = 0$

???

$-2/3$

2.



Braille letter

Quadratic Equation and Function, group *Omicron o* -iii

Quadratic Equation and Function, group *Omicron o* -iv

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a)

$8x^2 - 2x - 1 = 0$

...

???

...

1
- (b)

$-7x^2 + 8x + 7 = 0$

..

???

..

2
- (c)

$f(x) = 8x^2 - x - 8$

???

$-1/16$
- (d)

$f(x) = 3x^2 + 6x - 9$

???

$-15/2$
- (e)

$-2x^2 - 16x - 14 = 0$

???

-6
- (f)

$x^2 + 3x + 2 = 0$

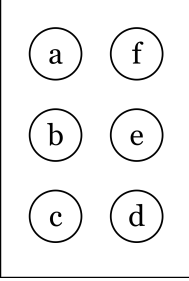
..

???

..

-3

3.



Braille letter

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a)

$-2x^2 - 4x - 7 = 0$

..

???

..

0
- (b)

$4x^2 + 9x - 4 = 0$

...

???

...

2
- (c)

$f(x) = -2x^2 + 2x - 3$

???

$1/2$
- (d)

$f(x) = x^2 + 5x - 2$

???

$-29/4$
- (e)

$-x^2 - 8x - 15 = 0$

.

???

.

-7
- (f)

$-4x^2 + 4x + 15 = 0$

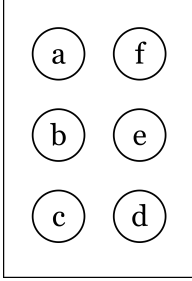
.

???

.

1

4.



Braille letter

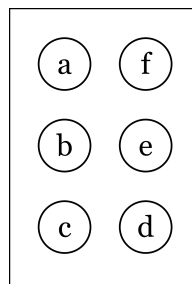
Quadratic Equation and Function, group $Pi \pi$ -i

Name:

- (a) Determine the number of real solutions. (d) Determine the y-coordinate of the vertex.
 (b) Determine the number of real solutions. (e) Calculate the sum of the solutions.
 (c) Determine the x-coordinate of the vertex. (f) Calculate the sum of the solutions.

- (a) $7x^2 - 3x + 9 = 0$.. ??? .. 1
 (b) $-7x^2 - x - 4 = 0$. ??? . 0
 (c) $f(x) = x^2 + x + 1$??? $-1/2$
 (d) $f(x) = x^2 - 4x + 1$??? $-7/2$
 (e) $-x^2 - 2x + 35 = 0$. ??? . -5
 (f) $3x^2 + 3x - 18 = 0$??? -1

1.



Braille letter

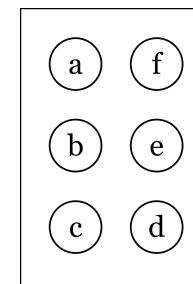
Quadratic Equation and Function, group $Pi \pi$ -ii

Name:

- (a) Determine the number of real solutions. (d) Determine the y-coordinate of the vertex.
 (b) Determine the number of real solutions. (e) Calculate the sum of the solutions.
 (c) Determine the x-coordinate of the vertex. (f) Calculate the sum of the solutions.

- (a) $5x^2 - x - 7 = 0$... ??? ... 2
 (b) $3x^2 - x + 5 = 0$... ??? ... 1
 (c) $f(x) = -2x^2 + x - 8$??? $-1/4$
 (d) $f(x) = 2x^2 + 4x + 1$??? $-3/2$
 (e) $-x^2 - 4x - 3 = 0$.. ??? .. -6
 (f) $2x^2 + 14x + 24 = 0$.. ??? .. 1

2.



Braille letter

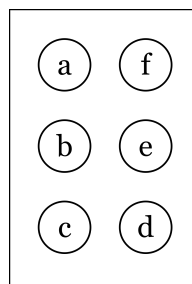
Quadratic Equation and Function, group $Pi \pi$ -iii

Name:

- (a) Determine the number of real solutions. (d) Determine the y-coordinate of the vertex.
 (b) Determine the number of real solutions. (e) Calculate the sum of the solutions.
 (c) Determine the x-coordinate of the vertex. (f) Calculate the sum of the solutions.

- (a) $2x^2 + 7x - 7 = 0$... ??? ... 2
 (b) $-5x^2 - 2x + 8 = 0$.. ??? .. 2
 (c) $f(x) = -x^2 + 4x - 1$. ??? . 2
 (d) $f(x) = -7x^2 - 2x - 1$??? $-6/7$
 (e) $5x^2 + 15x + 10 = 0$.. ??? .. 0
 (f) $2x^2 - 4x - 16 = 0$... ??? ... 6

3.



Braille letter

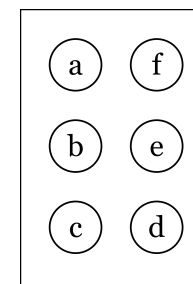
Quadratic Equation and Function, group $Pi \pi$ -iv

Name:

- (a) Determine the number of real solutions. (d) Determine the y-coordinate of the vertex.
 (b) Determine the number of real solutions. (e) Calculate the sum of the solutions.
 (c) Determine the x-coordinate of the vertex. (f) Calculate the sum of the solutions.

- (a) $-3x^2 - 6x - 5 = 0$.. ??? .. 0
 (b) $7x^2 + 3x - 2 = 0$... ??? ... 1
 (c) $f(x) = 2x^2 + 2x - 3$. ??? . $1/2$
 (d) $f(x) = 4x^2 - 3x - 1$??? $-17/16$
 (e) $-4x^2 + 28x - 48 = 0$. ??? . 7
 (f) $5x^2 + 5x - 10 = 0$.. ??? .. -3

4.



Braille letter

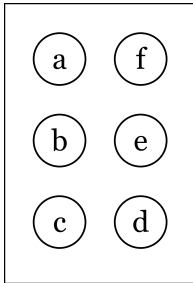
Quadratic Equation and Function, group *Rho* ρ -i

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $-2x^2 + 5x + 1 = 0$.. ??? .. 1
- (b) $2x^2 - x - 1 = 0$??? 2
- (c) $f(x) = 2x^2 + 6x - 7$??? $-3/2$
- (d) $f(x) = -4x^2 + 4x - 7$??? $-5/2$
- (e) $x^2 - 2x - 3 = 0$??? 0
- (f) $7x^2 - 6x - 1 = 0$... ??? ... $6/7$

1.



Braille letter

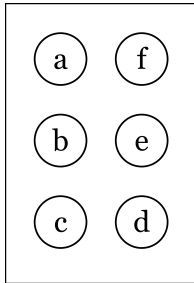
Quadratic Equation and Function, group *Rho* ρ -ii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $9x^2 + 6x + 5 = 0$??? 0
- (b) $4x^2 - 5x - 4 = 0$??? 0
- (c) $f(x) = -4x^2 + 6x - 3$??? $-3/4$
- (d) $f(x) = -6x^2 - x - 1$??? $-11/24$
- (e) $3x^2 - 21x + 30 = 0$.. ??? .. 5
- (f) $12x^2 - x - 6 = 0$. ??? . $-17/12$

2.



Braille letter

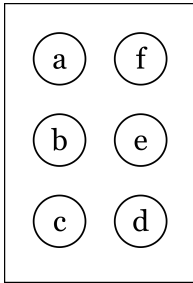
Quadratic Equation and Function, group *Rho* ρ -iii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $-x^2 - 7x - 1 = 0$.. ??? .. 2
- (b) $8x^2 + x - 2 = 0$... ??? ... 2
- (c) $f(x) = x^2 - 5x - 3$??? $5/2$
- (d) $f(x) = 3x^2 + 2x + 6$??? $8/3$
- (e) $x^2 + 2x - 8 = 0$... ??? ... 0
- (f) $-5x^2 - 4x + 12 = 0$??? $16/5$

3.



Braille letter

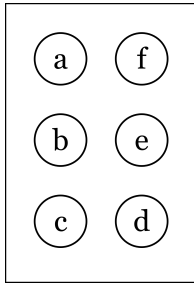
Quadratic Equation and Function, group *Rho* ρ -iv

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $x^2 - x + 7 = 0$??? 0
- (b) $4x^2 - 3x - 7 = 0$... ??? ... 1
- (c) $f(x) = 2x^2 - 3x - 4$??? $-3/4$
- (d) $f(x) = 4x^2 - 7x - 2$??? $-65/16$
- (e) $x^2 + 4x + 3 = 0$... ??? ... -4
- (f) $2x^2 - 13x + 20 = 0$. ??? . $-3/2$

4.



Braille letter

Quadratic Equation and Function, group *Sigma* σ -i

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $4x^2 - 6x + 4 = 0$... ??? ... 0

(b) $-4x^2 - 4x + 2 = 0$.. ??? .. 2

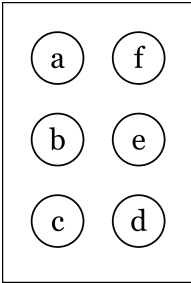
(c) $f(x) = -9x^2 - 3x - 4$??? $1/6$

(d) $f(x) = 3x^2 - 5x + 2$??? $-13/12$

(e) $-3x^2 + 12x - 9 = 0$.. ??? .. 4

(f) $6x^2 + 8x + 2 = 0$... ??? ... $2/3$

1.



Braille letter

Quadratic Equation and Function, group *Sigma* σ -ii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $3x^2 + 5x + 3 = 0$... ??? ... 0

(b) $x^2 - 5x + 2 = 0$... ??? ... 0

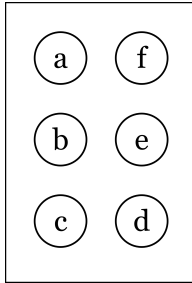
(c) $f(x) = 6x^2 + x + 1$. ??? . $1/12$

(d) $f(x) = -x^2 + 2x - 7$??? $-5/2$

(e) $4x^2 + 4x - 24 = 0$.. ??? .. 0

(f) $-10x^2 + 14x - 4 = 0$??? $-3/5$

2.



Braille letter

Quadratic Equation and Function, group *Sigma* σ -iii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-2x^2 + 5x + 1 = 0$.. ??? .. 2

(b) $-x^2 + 3x + 1 = 0$.. ??? .. 2

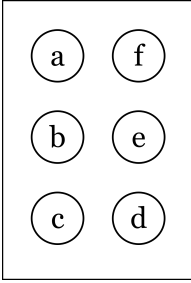
(c) $f(x) = -3x^2 + 2x - 6$??? $1/3$

(d) $f(x) = -x^2 + x - 7$??? $-13/4$

(e) $x^2 - 9x + 14 = 0$.. ??? .. 11

(f) $3x^2 - 15x + 18 = 0$. ??? . -1

3.



Braille letter

Quadratic Equation and Function, group *Sigma* σ -iv

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-6x^2 + x - 3 = 0$... ??? ... 0

(b) $-6x^2 - x + 7 = 0$... ??? ... 2

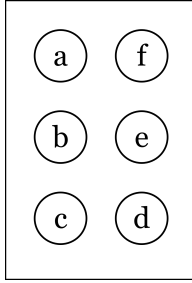
(c) $f(x) = -8x^2 - x - 2$??? $1/16$

(d) $f(x) = 4x^2 - 5x + 1$??? $-17/16$

(e) $-x^2 - 3x + 10 = 0$.. ??? .. -6

(f) $-4x^2 + 2x + 2 = 0$.. ??? .. $1/2$

4.



Braille letter

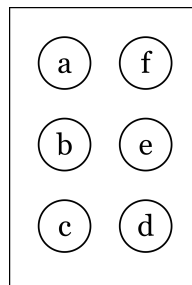
Quadratic Equation and Function, group *Tau* τ -i

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $-x^2 - 6x + 4 = 0$.. ??? .. 2
- (b) $2x^2 + 3x - 4 = 0$... ??? ... 2
- (c) $f(x) = 8x^2 - 9x + 4$??? $-9/16$
- (d) $f(x) = 4x^2 + x + 7$??? $55/16$
- (e) $4x^2 - 28x + 40 = 0$.. ??? .. 7
- (f) $5x^2 + 9x + 4 = 0$.. ??? .. $-1/5$

1.



Braille letter

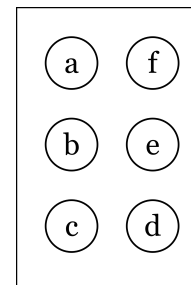
Quadratic Equation and Function, group *Tau* τ -ii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $x^2 - 5x - 6 = 0$??? 2
- (b) $3x^2 - 7x + 8 = 0$??? 2
- (c) $f(x) = -2x^2 + x + 7$??? $-1/4$
- (d) $f(x) = -5x^2 + 4x - 3$??? $-7/10$
- (e) $2x^2 + 10x + 12 = 0$.. ??? .. -5
- (f) $2x^2 + 3x + 1 = 0$... ??? ... $1/2$

2.



Braille letter

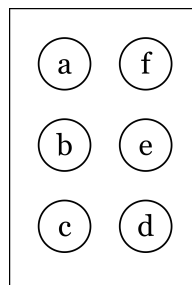
Quadratic Equation and Function, group *Tau* τ -iii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $3x^2 - 3x + 6 = 0$... ??? ... 0
- (b) $-5x^2 - 8x - 3 = 0$.. ??? .. 1
- (c) $f(x) = -8x^2 - 4x + 1$??? $1/4$
- (d) $f(x) = 3x^2 - 4x + 4$??? $2/3$
- (e) $9x^2 - 18x - 27 = 0$.. ??? .. 5
- (f) $8x^2 + 2x - 6 = 0$. ??? . $-7/4$

3.



Braille letter

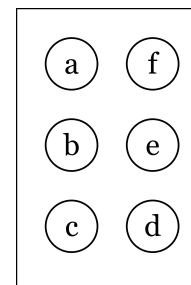
Quadratic Equation and Function, group *Tau* τ -iv

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

- (a) $-3x^2 + 5x - 9 = 0$.. ??? .. 0
- (b) $6x^2 - 9x - 4 = 0$... ??? ... 0
- (c) $f(x) = -6x^2 + 3x + 9$??? $-1/4$
- (d) $f(x) = -x^2 + 6x - 3$??? $15/2$
- (e) $5x^2 - 25x + 20 = 0$.. ??? .. 5
- (f) $-2x^2 + 9x - 4 = 0$.. ??? .. $9/2$

4.



Braille letter

Quadratic Equation and Function, group *Upsilon v* -i

Quadratic Equation and Function, group *Upsilon v* -ii

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $x^2 - 6x - 4 = 0$??? 2

(b) $3x^2 + 3x - 3 = 0$... ??? ... 2

(c) $f(x) = -7x^2 - 4x - 4$??? $-2/7$

(d) $f(x) = -4x^2 + 8x + 2$. ??? . 6

(e) $5x^2 - 5x - 10 = 0$... ??? ... 1

(f) $10x^2 - 6x - 4 = 0$.. ??? .. $7/5$

1.

a

f

b

e

c

d

Braille letter

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $-9x^2 - 6x - 8 = 0$. ??? . 0

(b) $6x^2 - 9x - 2 = 0$.. ??? .. 0

(c) $f(x) = -x^2 - 2x - 5$??? 1

(d) $f(x) = 2x^2 + 2x + 2$??? $1/2$

(e) $-5x^2 + 25x - 20 = 0$??? 2

(f) $10x^2 + 11x + 3 = 0$??? $1/10$

2.

a

f

b

e

c

d

Braille letter

Quadratic Equation and Function, group *Upsilon v* -iii

Quadratic Equation and Function, group *Upsilon v* -iv

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $2x^2 + 3x - 3 = 0$... ??? ... 2

(b) $-6x^2 + 8x - 7 = 0$.. ??? .. 0

(c) $f(x) = 3x^2 - 2x - 5$. ??? . $1/3$

(d) $f(x) = -2x^2 + 3x + 5$??? $29/8$

(e) $2x^2 - 10x - 28 = 0$.. ??? .. 2

(f) $6x^2 + 8x + 2 = 0$.. ??? .. $-2/3$

3.

a

f

b

e

c

d

Braille letter

Name:

(a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.

(d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

(a) $3x^2 + 6x + 4 = 0$.. ??? .. 0

(b) $-6x^2 + 2x + 1 = 0$. ??? . 0

(c) $f(x) = x^2 + x - 3$??? $-1/2$

(d) $f(x) = 7x^2 + 6x + 4$??? $5/7$

(e) $-x^2 - 7x - 6 = 0$. ??? . -9

(f) $-9x^2 + 12x - 3 = 0$??? $2/3$

4.

a

f

b

e

c

d

Braille letter

Quadratic Equation and Function, group *Phi* ϕ -i

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $6x^2 + 2x - 2 = 0$.. ??? .. 2

(b) $-3x^2 + x - 9 = 0$.. ??? .. 1

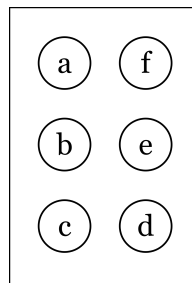
(c) $f(x) = 3x^2 + 4x + 4$??? $-2/3$

(d) $f(x) = 3x^2 - x + 4$??? $23/12$

(e) $3x^2 + 21x + 30 = 0$??? -10

(f) $3x^2 - 8x + 5 = 0$.. ??? .. $8/3$

1.



Braille letter

Quadratic Equation and Function, group *Phi* ϕ -ii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $3x^2 + 4x + 9 = 0$.. ??? .. 0

(b) $-3x^2 + 6x - 3 = 0$.. ??? .. 2

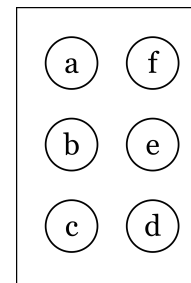
(c) $f(x) = 8x^2 - 6x - 1$??? $-3/8$

(d) $f(x) = 2x^2 - 3x + 6$??? $15/8$

(e) $3x^2 + 6x - 45 = 0$. ??? . -4

(f) $-2x^2 + 12x - 16 = 0$??? -2

2.



Braille letter

Quadratic Equation and Function, group *Phi* ϕ -iii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $-8x^2 + 3x + 3 = 0$.. ??? .. 1

(b) $3x^2 + 6x + 5 = 0$... ??? ... 0

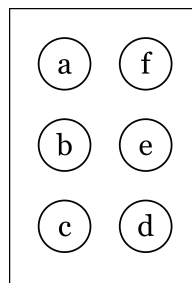
(c) $f(x) = 4x^2 - x + 3$. ??? . $-1/8$

(d) $f(x) = -3x^2 + 2x - 4$??? $-5/3$

(e) $3x^2 + 9x + 6 = 0$... ??? ... 0

(f) $-2x^2 + 14x - 20 = 0$. ??? . 7

3.



Braille letter

Quadratic Equation and Function, group *Phi* ϕ -iv

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $2x^2 + 5x - 7 = 0$.. ??? .. 2

(b) $4x^2 - 3x + 5 = 0$.. ??? .. 0

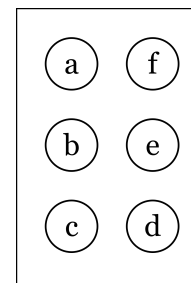
(c) $f(x) = -x^2 - 6x + 3$??? -3

(d) $f(x) = -2x^2 + 6x - 5$??? 2

(e) $4x^2 - 4x - 48 = 0$.. ??? .. 3

(f) $2x^2 - 6x + 4 = 0$. ??? . -1

4.



Braille letter

Quadratic Equation and Function, group *Chi* χ -i

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $2x^2 + 2x + 3 = 0$... ??? ... 1

(b) $-8x^2 - 5x - 7 = 0$.. ??? .. 0

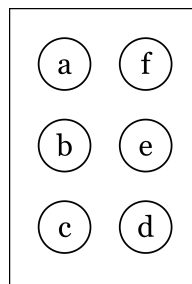
(c) $f(x) = -7x^2 - 2x + 4$??? $-1/7$

(d) $f(x) = 3x^2 - 4x + 4$. ??? . $2/3$

(e) $2x^2 + 10x - 28 = 0$.. ??? .. -2

(f) $-14x^2 + x + 4 = 0$. ??? . $1/14$

1.



Braille letter

Quadratic Equation and Function, group *Chi* χ -ii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $7x^2 + 6x + 5 = 0$... ??? ... 0

(b) $3x^2 - 5x - 3 = 0$... ??? ... 1

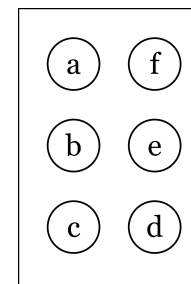
(c) $f(x) = -3x^2 - 9x - 1$??? $3/2$

(d) $f(x) = 2x^2 - 8x - 3$??? $-19/2$

(e) $-6x^2 - 6x + 72 = 0$. ??? . -1

(f) $x^2 - 3x - 18 = 0$... ??? ... 9

2.



Braille letter

Quadratic Equation and Function, group *Chi* χ -iii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $-7x^2 - 5x + 7 = 0$.. ??? .. 2

(b) $-5x^2 + 3x - 4 = 0$.. ??? .. 2

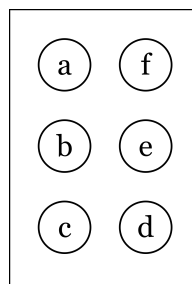
(c) $f(x) = 3x^2 - 3x - 1$??? $-1/2$

(d) $f(x) = -x^2 + x + 5$??? $11/4$

(e) $-4x^2 + 8x + 32 = 0$. ??? . 3

(f) $-2x^2 + 4x + 30 = 0$. ??? . 8

3.



Braille letter

Quadratic Equation and Function, group *Chi* χ -iv

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $-5x^2 + 3x + 2 = 0$... ??? ... 0

(b) $-x^2 - 5x + 5 = 0$??? 2

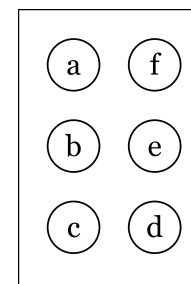
(c) $f(x) = x^2 + 5x + 1$.. ??? .. $-5/2$

(d) $f(x) = -6x^2 + x - 9$??? $-107/24$

(e) $-2x^2 + 8x - 6 = 0$... ??? ... 4

(f) $-15x^2 - 13x - 2 = 0$??? $-13/15$

4.



Braille letter

Quadratic Equation and Function, group *Psi* ψ -i

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $5x^2 + 3x + 2 = 0$... ??? ... 0

(b) $-x^2 - 9x - 2 = 0$.. ??? .. 0

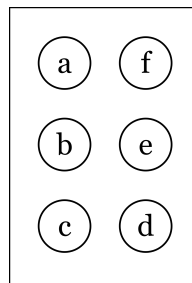
(c) $f(x) = -x^2 - x + 7$. ??? . $1/2$

(d) $f(x) = -7x^2 - 7x - 1$??? $5/4$

(e) $4x^2 + 8x - 12 = 0$.. ??? .. -2

(f) $-3x^2 + 6x + 9 = 0$.. ??? .. 2

1.



Braille letter

Quadratic Equation and Function, group *Psi* ψ -ii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $2x^2 - 3x + 4 = 0$... ??? ... 0

(b) $-2x^2 - 3x + 3 = 0$.. ??? .. 0

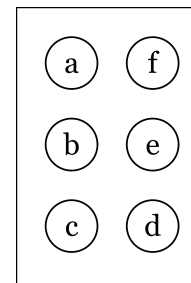
(c) $f(x) = 4x^2 + 8x - 2$. ??? . 1

(d) $f(x) = -6x^2 - 3x + 2$??? $11/8$

(e) $x^2 - 3x - 4 = 0$??? 3

(f) $-x^2 + x + 6 = 0$.. ??? .. -5

2.



Braille letter

Quadratic Equation and Function, group *Psi* ψ -iii

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $x^2 - 6x - 1 = 0$... ??? ... 0

(b) $-x^2 - x - 1 = 0$.. ??? .. 0

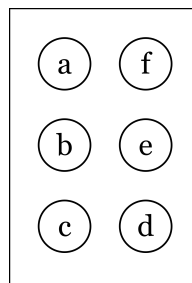
(c) $f(x) = -2x^2 + x - 7$??? $1/4$

(d) $f(x) = 2x^2 - x + 2$??? $7/8$

(e) $x^2 + 11x + 30 = 0$. ??? . -13

(f) $-3x^2 - 7x - 4 = 0$??? $-7/3$

3.



Braille letter

Quadratic Equation and Function, group *Psi* ψ -iv

Name:

- | | |
|---|---|
| (a) Determine the number of real solutions. | (d) Determine the y-coordinate of the vertex. |
| (b) Determine the number of real solutions. | (e) Calculate the sum of the solutions. |
| (c) Determine the x-coordinate of the vertex. | (f) Calculate the sum of the solutions. |

(a) $-x^2 - 4x + 4 = 0$... ??? ... 2

(b) $2x^2 - 2x + 8 = 0$... ??? ... 2

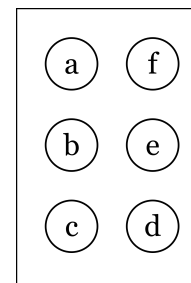
(c) $f(x) = -3x^2 - 2x + 5$??? $-1/3$

(d) $f(x) = -8x^2 + x + 1$??? $17/32$

(e) $-x^2 - 9x - 8 = 0$.. ??? .. -11

(f) $6x^2 + 11x + 4 = 0$.. ??? .. $5/6$

4.



Braille letter

Quadratic Equation and Function, group *Omega ω* -i

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $4x^2 + 3x - 7 = 0$... ??? ... 2

(b) $3x^2 + 9x + 1 = 0$... ??? ... 2

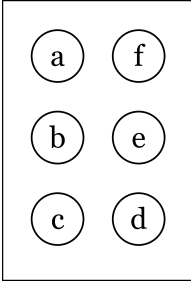
(c) $f(x) = -6x^2 + x + 2$??? $-1/12$

(d) $f(x) = 2x^2 + x + 4$. ??? . $15/8$

(e) $-x^2 - x + 20 = 0$... ??? ... 2

(f) $2x^2 + x - 21 = 0$.. ??? .. $-1/2$

1.



Braille letter

Quadratic Equation and Function, group *Omega ω* -ii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-3x^2 - 7x - 7 = 0$. ??? . 0

(b) $-5x^2 - 5x - 7 = 0$. ??? . 1

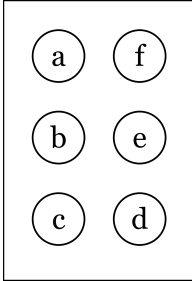
(c) $f(x) = 2x^2 - 2x + 2$??? $1/2$

(d) $f(x) = x^2 + 7x - 6$??? $-61/4$

(e) $8x^2 - 32x + 24 = 0$. ??? . 4

(f) $5x^2 + 8x - 4 = 0$. ??? . $12/5$

2.



Braille letter

Quadratic Equation and Function, group *Omega ω* -iii

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-3x^2 - 4x + 7 = 0$... ??? ... 2

(b) $-x^2 + 7x + 1 = 0$... ??? ... 1

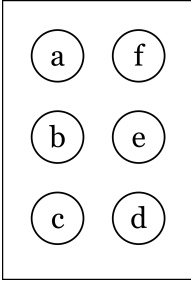
(c) $f(x) = -3x^2 - x + 6$??? $-1/6$

(d) $f(x) = -9x^2 - x - 4$??? $-71/36$

(e) $-9x^2 + 45x - 54 = 0$.. ??? .. 5

(f) $30x^2 + 4x - 2 = 0$. ??? . $-8/15$

3.



Braille letter

Quadratic Equation and Function, group *Omega ω* -iv

Name:

- (a) Determine the number of real solutions.

(b) Determine the number of real solutions.

(c) Determine the x-coordinate of the vertex.
- (d) Determine the y-coordinate of the vertex.

(e) Calculate the sum of the solutions.

(f) Calculate the sum of the solutions.

- (a) $-4x^2 - 6x + 2 = 0$.. ??? .. 2

(b) $-3x^2 + 2x + 4 = 0$.. ??? .. 1

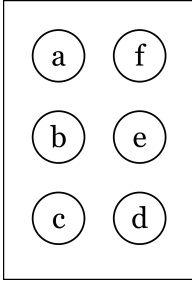
(c) $f(x) = -x^2 - 6x + 3$. ??? . 3

(d) $f(x) = -6x^2 + x + 2$??? $25/24$

(e) $-5x^2 - 5x + 30 = 0$. ??? . -1

(f) $5x^2 - 9x - 2 = 0$.. ??? .. $9/5$

4.



Braille letter

Quadratic Equation and Function (solutions)

α	$i : \mathbf{T}$	(a) 0 $\times$	(b) 0 $\checkmark$	(c) $2/7\checkmark$	(d) $28/3\times$	(e) $-6\checkmark$	(f) $1\checkmark$
	$ii : \mathbf{O}$	(a) 0 $\checkmark$	(b) 2 $\times$	(c) $2/3\checkmark$	(d) 8 $\times$	(e) $-1\checkmark$	(f) $3/10\times$
	$iii : \mathbf{W}$	(a) 2 $\checkmark$	(b) 2 $\checkmark$	(c) $2/7\checkmark$	(d) $-23/8\checkmark$	(e) $-3\checkmark$	(f) $-5/12\times$
	$iv : \mathbf{N}$	(a) 0 $\checkmark$	(b) 2 $\times$	(c) $-1/2\checkmark$	(d) 3 $\times$	(e) $-5\checkmark$	(f) $-1/2\checkmark$
β	$i : \mathbf{U}$	(a) 0 $\checkmark$	(b) 0 $\times$	(c) $-1/4\checkmark$	(d) 22 $\checkmark$	(e) $-1\times$	(f) $1/2\times$
	$ii : \mathbf{N}$	(a) 0 $\checkmark$	(b) 0 $\times$	(c) $-1\checkmark$	(d) $85/28\times$	(e) $-4\checkmark$	(f) $-1/6\checkmark$
	$iii : \mathbf{I}$	(a) 2 $\times$	(b) 2 $\checkmark$	(c) $1/6\times$	(d) $7/4\times$	(e) 7 $\times$	(f) $6/5\checkmark$
	$iv : \mathbf{T}$	(a) 0 $\times$	(b) 2 $\checkmark$	(c) $-3/4\checkmark$	(d) $33/8\times$	(e) $-11\checkmark$	(f) $5/4\checkmark$
γ	$i : \mathbf{W}$	(a) 2 $\checkmark$	(b) 2 $\checkmark$	(c) $3/10\checkmark$	(d) $-13/12\checkmark$	(e) 10 $\checkmark$	(f) $-11/2\times$
	$ii : \mathbf{A}$	(a) 2 $\checkmark$	(b) 2 $\times$	(c) $5/4\times$	(d) $1/2\times$	(e) 2 $\times$	(f) $7/15\times$
	$iii : \mathbf{L}$	(a) 2 $\checkmark$	(b) 2 $\checkmark$	(c) $1/4\checkmark$	(d) $-31/16\times$	(e) $-4\times$	(f) $-13/3\times$
	$iv : \mathbf{L}$	(a) 2 $\checkmark$	(b) 2 $\checkmark$	(c) $-2\checkmark$	(d) $-49/4\times$	(e) 2 $\times$	(f) $-9/5\times$
δ	$i : \mathbf{F}$	(a) 0 $\checkmark$	(b) 0 $\checkmark$	(c) $3/8\times$	(d) $-14/3\times$	(e) 2 $\times$	(f) $-3/4\checkmark$
	$ii : \mathbf{A}$	(a) 0 $\checkmark$	(b) 0 $\times$	(c) $-3/10\times$	(d) $-1/8\times$	(e) $-2\times$	(f) $7/4\times$
	$iii : \mathbf{R}$	(a) 0 $\checkmark$	(b) 0 $\checkmark$	(c) $-1/7\checkmark$	(d) $23/8\times$	(e) 7 $\checkmark$	(f) 1 $\times$
	$iv : \mathbf{M}$	(a) 2 $\checkmark$	(b) 2 $\times$	(c) $-1/5\checkmark$	(d) $-2\times$	(e) 9 $\times$	(f) $5/6\checkmark$
ϵ	$i : \mathbf{E}$	(a) 0 $\checkmark$	(b) 0 $\times$	(c) $-1/14\times$	(d) $59/28\times$	(e) 5 $\checkmark$	(f) $-7/6\times$
	$ii : \mathbf{A}$	(a) 0 $\checkmark$	(b) 2 $\times$	(c) $-2/3\times$	(d) $7/4\times$	(e) $-3\times$	(f) 10 $\times$
	$iii : \mathbf{S}$	(a) 2 $\times$	(b) 0 $\checkmark$	(c) $7/10\checkmark$	(d) $-21/4\times$	(e) 1 $\times$	(f) $-5\checkmark$
	$iv : \mathbf{T}$	(a) 0 $\times$	(b) 0 $\checkmark$	(c) $-3/2\checkmark$	(d) $-23/4\times$	(e) $-1\checkmark$	(f) 3 $\checkmark$
ζ	$i : \mathbf{M}$	(a) 2 $\checkmark$	(b) 2 $\times$	(c) $9/10\checkmark$	(d) $-3/4\times$	(e) 6 $\times$	(f) $3/4\checkmark$
	$ii : \mathbf{I}$	(a) 2 $\times$	(b) 0 $\checkmark$	(c) $-4/3\times$	(d) $65/4\times$	(e) $-9\times$	(f) $-5/3\checkmark$
	$iii : \mathbf{N}$	(a) 0 $\checkmark$	(b) 2 $\times$	(c) $7/8\checkmark$	(d) $47/8\times$	(e) $-3\checkmark$	(f) $-2/15\checkmark$
	$iv : \mathbf{D}$	(a) 2 $\checkmark$	(b) 2 $\times$	(c) $-1\times$	(d) $15/16\times$	(e) $-3\checkmark$	(f) $-2\checkmark$
η	$i : \mathbf{S}$	(a) 1 $\times$	(b) 2 $\checkmark$	(c) $-2\checkmark$	(d) $29/4\times$	(e) 3 $\times$	(f) $1/2\checkmark$
	$ii : \mathbf{I}$	(a) 0 $\times$	(b) 0 $\checkmark$	(c) $-1/18\times$	(d) $11/5\times$	(e) $-5\times$	(f) $5/2\checkmark$
	$iii : \mathbf{Z}$	(a) 0 $\checkmark$	(b) 2 $\times$	(c) $1/16\checkmark$	(d) $-9\checkmark$	(e) 5 $\checkmark$	(f) $-7/12\times$
	$iv : \mathbf{E}$	(a) 2 $\checkmark$	(b) 2 $\times$	(c) $-1/5\times$	(d) $33/7\times$	(e) $-9\checkmark$	(f) $1/6\times$
θ	$i : \mathbf{D}$	(a) 2 $\checkmark$	(b) 0 $\times$	(c) $-1/4\times$	(d) $-127/16\times$	(e) 7 $\checkmark$	(f) 5 $\checkmark$
	$ii : \mathbf{I}$	(a) 0 $\times$	(b) 2 $\checkmark$	(c) $-5/6\times$	(d) $45/4\times$	(e) $-7\times$	(f) $-1\checkmark$
	$iii : \mathbf{R}$	(a) 2 $\checkmark$	(b) 2 $\checkmark$	(c) $9/8\checkmark$	(d) $13/3\times$	(e) 4 $\checkmark$	(f) $3/4\times$
	$iv : \mathbf{T}$	(a) 2 $\times$	(b) 2 $\checkmark$	(c) 2 $\checkmark$	(d) $-26/5\times$	(e) 2 $\checkmark$	(f) $2/3\checkmark$
ι	$i : \mathbf{B}$	(a) 2 $\checkmark$	(b) 2 $\checkmark$	(c) $-7/12\times$	(d) $-19/4\times$	(e) $-2\times$	(f) $-2\times$
	$ii : \mathbf{O}$	(a) 2 $\checkmark$	(b) 0 $\times$	(c) $5/14\checkmark$	(d) $5/8\times$	(e) $-9\checkmark$	(f) $-7/2\times$
	$iii : \mathbf{S}$	(a) 0 $\times$	(b) 2 $\checkmark$	(c) $1/3\checkmark$	(d) $40/9\times$	(e) $-2\times$	(f) $-3/2\checkmark$
	$iv : \mathbf{S}$	(a) 0 $\times$	(b) 2 $\checkmark$	(c) $3/8\checkmark$	(d) $265/32\times$	(e) $-1\times$	(f) $1/3\checkmark$
κ	$i : \mathbf{G}$	(a) 2 $\checkmark$	(b) 0 $\checkmark$	(c) $-4/3\times$	(d) $53/8\times$	(e) 5 $\checkmark$	(f) $-1/2\checkmark$
	$ii : \mathbf{A}$	(a) 2 $\checkmark$	(b) 1 $\times$	(c) $1/2\times$	(d) $-1/4\times$	(e) $-9\times$	(f) $-15/8\times$
	$iii : \mathbf{T}$	(a) 2 $\times$	(b) 2 $\checkmark$	(c) $-5/6\checkmark$	(d) 6 $\times$	(e) $-12\checkmark$	(f) 1 $\checkmark$
	$iv : \mathbf{E}$	(a) 2 $\checkmark$	(b) 0 $\times$	(c) $7/16\times$	(d) $-73/24\times$	(e) 3 $\checkmark$	(f) $-1/2\times$
λ	$i : \mathbf{F}$	(a) 0 $\checkmark$	(b) 0 $\checkmark$	(c) $4/3\times$	(d) $19/4\times$	(e) $-10\times$	(f) $-3\checkmark$
	$ii : \mathbf{I}$	(a) 2 $\times$	(b) 2 $\checkmark$	(c) 4 $\times$	(d) $89/16\times$	(e) 10 $\times$	(f) $1/4\checkmark$
	$iii : \mathbf{V}$	(a) 0 $\checkmark$	(b) 0 $\checkmark$	(c) $1/18\checkmark$	(d) $-5\checkmark$	(e) 9 $\times$	(f) $-17/12\times$
	$iv : \mathbf{E}$	(a) 2 $\checkmark$	(b) 2 $\times$	(c) $-1/4\times$	(d) $-33/4\times$	(e) $-1\checkmark$	(f) $-5/3\times$
μ	$i : \mathbf{W}$	(a) 0 $\checkmark$	(b) 2 $\checkmark$	(c) $-9/4\checkmark$	(d) $-5/4\checkmark$	(e) 3 $\checkmark$	(f) $-5/2\times$
	$ii : \mathbf{O}$	(a) 0 $\checkmark$	(b) 2 $\times$	(c) $-1/10\checkmark$	(d) $177/16\times$	(e) 2 $\checkmark$	(f) $-7/3\times$
	$iii : \mathbf{O}$	(a) 2 $\checkmark$	(b) 0 $\times$	(c) $1/2\checkmark$	(d) $41/4\times$	(e) 2 $\checkmark$	(f) $5/12\times$
	$iv : \mathbf{D}$	(a) 2 $\checkmark$	(b) 2 $\times$	(c) 1 $\times$	(d) $29/4\times$	(e) $-6\checkmark$	(f) $4/3\checkmark$

Quadratic Equation and Function (solutions)

$\boxed{\nu}$	$i : \mathbf{R}$	(a) 0✓	(b) 0✓	(c) $7/10✓$	(d) $-37/12\mathbf{X}$	(e) 3✓	(f) 3 $\mathbf{X}$
	$ii : \mathbf{O}$	(a) 2✓	(b) 2 $\mathbf{X}$	(c) $1/9✓$	(d) $7/8\mathbf{X}$	(e) 1✓	(f) $3/14\mathbf{X}$
	$iii : \mathbf{A}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $-3/4\mathbf{X}$	(d) $95/12\mathbf{X}$	(e) $-7\mathbf{X}$	(f) $-5/2\mathbf{X}$
	$iv : \mathbf{D}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $1/16\mathbf{X}$	(d) $-11\mathbf{X}$	(e) $-6✓$	(f) $-4/7✓$
$\boxed{\xi}$	$i : \mathbf{G}$	(a) 2✓	(b) 0✓	(c) $-1/14\mathbf{X}$	(d) $-227/36\mathbf{X}$	(e) $-4✓$	(f) $1/6✓$
	$ii : \mathbf{R}$	(a) 2✓	(b) 2✓	(c) 1✓	(d) $-10/7\mathbf{X}$	(e) 4✓	(f) $-5/12\mathbf{X}$
	$iii : \mathbf{A}$	(a) 2✓	(b) 2 $\mathbf{X}$	(c) $7/6\mathbf{X}$	(d) $47/8\mathbf{X}$	(e) 2 $\mathbf{X}$	(f) 1 $\mathbf{X}$
	$iv : \mathbf{Y}$	(a) 2✓	(b) 2 $\mathbf{X}$	(c) $-3✓$	(d) $69/20✓$	(e) $-2✓$	(f) $1/2✓$
$\boxed{o}$	$i : \mathbf{C}$	(a) 0✓	(b) 0 $\mathbf{X}$	(c) $5/6\mathbf{X}$	(d) $-121/12\mathbf{X}$	(e) $-14\mathbf{X}$	(f) $1/3✓$
	$ii : \mathbf{L}$	(a) 2✓	(b) 2✓	(c) $-3/10✓$	(d) $113/16\mathbf{X}$	(e) $-2\mathbf{X}$	(f) $10/3\mathbf{X}$
	$iii : \mathbf{I}$	(a) 2 $\mathbf{X}$	(b) 2✓	(c) $1/16\mathbf{X}$	(d) $-12\mathbf{X}$	(e) $-8\mathbf{X}$	(f) $-3✓$
	$iv : \mathbf{P}$	(a) 0✓	(b) 2✓	(c) $1/2✓$	(d) $-33/4\mathbf{X}$	(e) $-8\mathbf{X}$	(f) 1✓
$\boxed{\pi}$	$i : \mathbf{S}$	(a) 0 $\mathbf{X}$	(b) 0✓	(c) $-1/2✓$	(d) $-3\mathbf{X}$	(e) $-2\mathbf{X}$	(f) $-1✓$
	$ii : \mathbf{A}$	(a) 2✓	(b) 0 $\mathbf{X}$	(c) $1/4\mathbf{X}$	(d) $-1\mathbf{X}$	(e) $-4\mathbf{X}$	(f) $-7\mathbf{X}$
	$iii : \mathbf{V}$	(a) 2✓	(b) 2✓	(c) 2✓	(d) $-6/7✓$	(e) $-3\mathbf{X}$	(f) 2 $\mathbf{X}$
	$iv : \mathbf{E}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $-1/2\mathbf{X}$	(d) $-25/16\mathbf{X}$	(e) 7✓	(f) $-1\mathbf{X}$
$\boxed{\rho}$	$i : \mathbf{S}$	(a) 2 $\mathbf{X}$	(b) 2✓	(c) $-3/2✓$	(d) $-6\mathbf{X}$	(e) 2 $\mathbf{X}$	(f) $6/7✓$
	$ii : \mathbf{A}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $3/4\mathbf{X}$	(d) $-23/24\mathbf{X}$	(e) 7 $\mathbf{X}$	(f) $1/12\mathbf{X}$
	$iii : \mathbf{L}$	(a) 2✓	(b) 2✓	(c) $5/2✓$	(d) $17/3\mathbf{X}$	(e) $-2\mathbf{X}$	(f) $-4/5\mathbf{X}$
	$iv : \mathbf{E}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $3/4\mathbf{X}$	(d) $-81/16\mathbf{X}$	(e) $-4✓$	(f) $13/2\mathbf{X}$
$\boxed{\sigma}$	$i : \mathbf{H}$	(a) 0✓	(b) 2✓	(c) $-1/6\mathbf{X}$	(d) $-1/12\mathbf{X}$	(e) 4✓	(f) $-4/3\mathbf{X}$
	$ii : \mathbf{A}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $-1/12\mathbf{X}$	(d) $-6\mathbf{X}$	(e) $-1\mathbf{X}$	(f) $7/5\mathbf{X}$
	$iii : \mathbf{L}$	(a) 2✓	(b) 2✓	(c) $1/3✓$	(d) $-27/4\mathbf{X}$	(e) 9 $\mathbf{X}$	(f) 5 $\mathbf{X}$
	$iv : \mathbf{F}$	(a) 0✓	(b) 2✓	(c) $-1/16\mathbf{X}$	(d) $-9/16\mathbf{X}$	(e) $-3\mathbf{X}$	(f) $1/2✓$
$\boxed{\tau}$	$i : \mathbf{H}$	(a) 2✓	(b) 2✓	(c) $9/16\mathbf{X}$	(d) $111/16\mathbf{X}$	(e) 7✓	(f) $-9/5\mathbf{X}$
	$ii : \mathbf{E}$	(a) 2✓	(b) 0 $\mathbf{X}$	(c) $1/4\mathbf{X}$	(d) $-11/5\mathbf{X}$	(e) $-5✓$	(f) $-3/2\mathbf{X}$
	$iii : \mathbf{A}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $-1/4\mathbf{X}$	(d) $8/3\mathbf{X}$	(e) 2 $\mathbf{X}$	(f) $-1/4\mathbf{X}$
	$iv : \mathbf{D}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $1/4\mathbf{X}$	(d) 6 $\mathbf{X}$	(e) 5✓	(f) $9/2✓$
$\boxed{v}$	$i : \mathbf{W}$	(a) 2✓	(b) 2✓	(c) $-2/7✓$	(d) 6✓	(e) 1✓	(f) $3/5\mathbf{X}$
	$ii : \mathbf{A}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $-1\mathbf{X}$	(d) $3/2\mathbf{X}$	(e) 5 $\mathbf{X}$	(f) $-11/10\mathbf{X}$
	$iii : \mathbf{L}$	(a) 2✓	(b) 0✓	(c) $1/3✓$	(d) $49/8\mathbf{X}$	(e) 5 $\mathbf{X}$	(f) $-4/3\mathbf{X}$
	$iv : \mathbf{K}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $-1/2✓$	(d) $19/7\mathbf{X}$	(e) $-7\mathbf{X}$	(f) $4/3\mathbf{X}$
$\boxed{\phi}$	$i : \mathbf{M}$	(a) 2✓	(b) 0 $\mathbf{X}$	(c) $-2/3✓$	(d) $47/12\mathbf{X}$	(e) $-7\mathbf{X}$	(f) $8/3✓$
	$ii : \mathbf{A}$	(a) 0✓	(b) 1 $\mathbf{X}$	(c) $3/8\mathbf{X}$	(d) $39/8\mathbf{X}$	(e) $-2\mathbf{X}$	(f) 6 $\mathbf{X}$
	$iii : \mathbf{I}$	(a) 2 $\mathbf{X}$	(b) 0✓	(c) $1/8\mathbf{X}$	(d) $-11/3\mathbf{X}$	(e) $-3\mathbf{X}$	(f) 7✓
	$iv : \mathbf{L}$	(a) 2✓	(b) 0✓	(c) $-3✓$	(d) $-1/2\mathbf{X}$	(e) 1 $\mathbf{X}$	(f) 3 $\mathbf{X}$
$\boxed{\chi}$	$i : \mathbf{S}$	(a) 0 $\mathbf{X}$	(b) 0✓	(c) $-1/7✓$	(d) $8/3\mathbf{X}$	(e) $-5\mathbf{X}$	(f) $1/14✓$
	$ii : \mathbf{E}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $-3/2\mathbf{X}$	(d) $-11\mathbf{X}$	(e) $-1✓$	(f) 3 $\mathbf{X}$
	$iii : \mathbf{A}$	(a) 2✓	(b) 0 $\mathbf{X}$	(c) $1/2\mathbf{X}$	(d) $21/4\mathbf{X}$	(e) 2 $\mathbf{X}$	(f) 2 $\mathbf{X}$
	$iv : \mathbf{T}$	(a) 2 $\mathbf{X}$	(b) 2✓	(c) $-5/2✓$	(d) $-215/24\mathbf{X}$	(e) 4✓	(f) $-13/15✓$
$\boxed{\psi}$	$i : \mathbf{D}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $-1/2\mathbf{X}$	(d) $3/4\mathbf{X}$	(e) $-2✓$	(f) 2✓
	$ii : \mathbf{E}$	(a) 0✓	(b) 2 $\mathbf{X}$	(c) $-1\mathbf{X}$	(d) $19/8\mathbf{X}$	(e) 3✓	(f) 1 $\mathbf{X}$
	$iii : \mathbf{S}$	(a) 2 $\mathbf{X}$	(b) 0✓	(c) $1/4✓$	(d) $15/8\mathbf{X}$	(e) $-11\mathbf{X}$	(f) $-7/3✓$
	$iv : \mathbf{K}$	(a) 2✓	(b) 0 $\mathbf{X}$	(c) $-1/3✓$	(d) $33/32\mathbf{X}$	(e) $-9\mathbf{X}$	(f) $-11/6\mathbf{X}$
$\boxed{\omega}$	$i : \mathbf{F}$	(a) 2✓	(b) 2✓	(c) $1/12\mathbf{X}$	(d) $31/8\mathbf{X}$	(e) $-1\mathbf{X}$	(f) $-1/2✓$
	$ii : \mathbf{O}$	(a) 0✓	(b) 0 $\mathbf{X}$	(c) $1/2✓$	(d) $-73/4\mathbf{X}$	(e) 4✓	(f) $-8/5\mathbf{X}$
	$iii : \mathbf{O}$	(a) 2✓	(b) 2 $\mathbf{X}$	(c) $-1/6✓$	(d) $-143/36\mathbf{X}$	(e) 5✓	(f) $-2/15\mathbf{X}$
	$iv : \mathbf{D}$	(a) 2✓	(b) 2 $\mathbf{X}$	(c) $-3\mathbf{X}$	(d) $49/24\mathbf{X}$	(e) $-1✓$	(f) $9/5✓$